The Reliability Tax: When Thinking Makes Language Models Less Trustworthy

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Abstract

Reasoning-augmented language models are deployed by default for high-stakes queries, and how long a model "thinks" is now a user-facing knob. The prevailing assumption is that more thinking yields more trustworthy answers. We test it. We formalize the capability–reliability trade-off as a measurable *reliability tax*, and ask whether it is a law—and whether it is *predictable*. A black-box budget sweep over four task types measures accuracy and three reliability axes jointly (calibration, selective prediction, abstention-versus-fabrication). A first real run already shows the tax is *task-dependent* and *switch-triggered*: on false-premise items, enabling thinking raises fabrication from 3.8% to 21.3%, while mathematical reasoning is unaffected.

1 Background & Motivation

Reasoning-augmented language models (LRMs) spend a variable amount of compute "thinking" before answering (DeepSeek-AI et al., 2025), and this budget is increasingly exposed as a user-facing control (Muennighoff et al., 2025; Snell et al., 2024). The implicit contract is simple: *more thinking buys more trustworthy answers*. Practitioners act on it—raising the budget to "max" whenever an answer matters.

That contract is being contradicted from several directions. Reinforcement learning that raises accuracy degrades calibration (Wang et al., 2026; Jin et al., 2025); enhancing reasoning amplifies tool hallucination, even at inference time (Yin et al., 2025); "slower" models are measurably *less* truthful under misleading input (Fang et al., 2025); and reasoning models stay confidently wrong on a large fraction of false-premise queries (Sathyanathan et al., 2026). Crucially, the evidence is *single-axis*, *training-side*, and *descriptive*: it shows the trade-off *exists*, not whether it is a law, or—the property a deployment rule needs—whether it is *predictable* from the trace before a budget is committed.

2 Research Questions & Hypotheses

RQ. Does increasing a model's thinking budget systematically trade reliability for accuracy, and can the degradation be foreseen?

- **H1 (task-dependence).** The ratio of reliability loss to accuracy gain differs across task types by at least a factor of two.
- **H2 (predictability, primary).** A predictor trained on a held-out half decides, at low budget and *without* seeing the high-budget answer, whether continuing will help; the induced policy beats the best fixed budget in *usable accuracy* at matched cost.
- **H3 (negative control).** Shuffling the trace features removes H2's advantage—if not, the predictor exploits an artefact.

Each claim carries an explicit falsification condition, and the design is pre-registered on a held-out prompt split.

3 Preliminary Literature

The closest work varies the reasoning budget and measures *one* axis: Lacombe et al. (2025) show over-reasoning impairs confidence calibration; Zhou et al. (2026) characterize the diminishing—sometimes negative—marginal utility of extra tokens. Yin et al. (2025) prove the capability–reliability trade-off is method-agnostic, and Fang et al. (2025) establish inverse truthfulness scaling. Our study differs on three points: it measures three reliability axes *jointly* to test whether they degrade as one; it spans tasks and model families rather than one benchmark; and it asks whether the degradation is *predictable*, turning the trade-off curve into a black-box budget rule. (40 references collected

and verified; ACL 2026 landscape survey of 1,409 papers informs the framing.)

4 Method & Experimental Design

The object. Everything is read off the *budget–reliability profile* of a model on a task: accuracy and each reliability axis as a function of the thinking budget. We report against the *realized* reasoning-token count, since nominal budget knobs are not comparable across providers.

The tax. With every reliability score oriented so that higher is better, $\tau_r = (\partial \rho_r / \partial b)/(\partial \mathrm{acc}/\partial b)$ is the reliability given up per unit of accuracy gained. Because τ_r diverges where accuracy saturates, our primary summaries are bounded instead: the fixed-coverage risk $R(c)$ and the area under the risk–coverage curve (Geifman and El-Yaniv, 2017).

Three axes. (i) *Calibration*: ECE (Guo et al., 2017) and verbalized overconfidence (Kadavath et al., 2022; Tian et al., 2023). (ii) *Selective prediction*: $R(c)$ and AURC. (iii) *Abstention vs. fabrication* on false-premise items, classified as refute / hedge / endorse, with fabrication rate = endorse share (Sathyanathan et al., 2026; Kirichenko et al., 2025).

Tasks and budget. Four task types span where compute should help and where it can hurt: mathematical reasoning (GSM8K, MATH-500; verifiable), factual QA (NQ), multi-hop QA (HotpotQA; composition), and false-premise items built from SciFact claims whose evidence contradicts them (Wadden et al., 2020). The budget axis uses the provider's official controls—thinking on/off and reasoning-effort levels—not `max_tokens`, which would confound budget with truncation; realized reasoning tokens are logged per response. Four budgets × two prompt templates × k=4 samples per item, with prompt-clustered bootstrap intervals.

From profile to decision. The profile yields an oracle: the budget maximizing *usable accuracy*, i.e. accuracy at the coverage where ECE stays below a target ϵ. We then test whether that oracle is *findable* from low-budget features alone (realized length, verbalized confidence, self-consistency, hedging markers), with the H3 permutation control.

5 Preliminary Result

A first real run on `deepseek-flash` (1,280 calls, 0 errors; ∼¥7–13) already indicates the tax is *task-dependent* and *switch-triggered*:

Task	Acc. off→max	ECE off→max	Fabrication off→max
Math	.988→.950	.01→.04	—
Factual QA	.475→.550	.45→.39	—
Multi-hop QA	.463→.562	.46→.41	—
False-premise	.962→.800	.03→.13	**.038→.213**

On false-premise items—where there is *no* accuracy to gain—enabling thinking multiplies fabrication by 5.6× and breaks calibration, while mathematical reasoning is untouched. Notably the effect is a *step* at the thinking on/off boundary rather than a smooth function of effort level, which sharpens H1 into a stronger, more falsifiable claim.

6 Expected Outcomes

(1) A cross-task, cross-family characterization of whether the trade-off is a law or a set of local effects, and where it is worst. (2) A held-out, cost-matched evaluation of an oracle-free *stopping rule* that decides when to stop thinking. (3) A released protocol, prompts, and raw responses for reproducibility.

7 Timeline & Responsibilities

Single-member project (all roles by the author).

- **Wk 1–2** (done): landscape survey; protocol and code built; pipeline validated; first real run on DeepSeek.
- **Wk 3**: scale DeepSeek to 200 items × k=4; add Qwen3, GLM, Doubao for cross-family coverage.
- **Wk 4**: train and evaluate the stopping rule (H2/H3); statistics.
- **Wk 5**: final report, slides, and artifact release.

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